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Is the future of fluid power digital?

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Abstract

This article presents digital fluid power as a new branch of fluid power which offers high potential for innovative solutions. The variety of digital concepts is quite large and digital concepts have long been successfully applied in low-power applications. Research and development is now becoming more intensive, being undertaken by several research groups and also more and more in industry. First applications will be brought to industry soon. A successful application requires new components, a sound understanding of system and new control principles.

Keywords

Digital fluid power, fluid power trends, switching control

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This article is a response to Achten's Forward Look article 'Convicted to innovation in fluid power',[1] in which he urges more courageous innovation to enable fluid power to exploit its potential to contribute towards tackling major societal challenges. He argues that the fluid power community should abandon its defensive position towards electro-mechanical drive systems and elaborate its intrinsic capability to develop powerful innovative solutions.

Are digital approaches an answer?

At first glance, conventional fluid power systems seem to be continuous by nature, but even a displacement pump operates discontinuously by periodically switching between suction and delivery phases. An intrinsic feature of fluid power is the use of valves to change the system structure. Thus discontinuous, variable structure control was discussed in the 19th century,[2–4] and switching control was applied in Montgolfier's hydraulic ram in the 18th century.[5] Subsequently, Flügge-Lotz and Taylor identified major advantages of such control to be simplicity and better power handling.[6] Many modern low-power systems implement control by on–off valves, e.g. anti-lock brakes and common rail diesel injection systems in vehicles, and special cooling systems. The fundamental principle in these systems is to combine simple hydraulic hardware with advanced control techniques. Thus fluid power systems have historically combined continuous and discrete elements with the application of discrete control concepts.

The overwhelming success of digital concepts in information, communication and power electronics, as well as the analogy between electrical and fluid power systems, suggest that digital concepts should be beneficial in fluid power. There have been several attempts to transfer non d.c. concepts of electrical engineering to fluid power, like alternating-current technology,[7,8] hydraulic transformer concepts (e.g. Dantlgraber[9]) and switching control.[10] These studies and other unpublished works have employed standard components which are generally inadequate for digital technologies. This situation compares with power electronics, which relies on powerful semiconductor components (often called 'valves' by power electronics experts); there would have been no breakthrough of switching control (e.g. by pulse–width modulation) in electrical drive technology with tube amplifiers!

Since the beginning of this millennium, the application of digital concepts in fluid power has been promoted progressively. Research and development at a few universities and companies has addressed system concepts and advanced components. Without the development of fast switching and check valves,

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compact and high-response accumulators, digital cylinders and digital displacement pumps, many digital systems would not work. Furthermore, adequate modelling and simulation methods have been developed to understand the hydraulic processes in such systems and these provide the means for systematic analysis and optimization. Research results show that digital concepts can be realized in various ways: digital displacement pumps and motors, digital cylinders, parallel valve technology, switching control with and without energy saving, switching converters and combinations of continuous and digital concepts. More details about the state of the art can be found elsewhere.[11] In all cases, control is effected by switching valves. Digital fluid power offers several advantages compared with analogue technologies, i.e. higher efficiency, precision, redundancy, robustness, higher component standardization potential and a better realization of small system drives or actuators.

There is an ongoing series of annual workshops on digital fluid power which attracts high interest from both academia and industry. The potential users in industry request a quick transfer of prototypes into practical solutions. Digital fluid power's novelty and the positive connotation of the word 'digital' are good selling points. The fluid power industry, however, is cautious. Companies do not want to generate excessive expectations in view of some challenging problems which require additional research to ensure a solid solution.

So we return to the crucial question, is the future of fluid power digital?

Fluid power is already today partially digital, but the current authors are convinced that it will become much more digital in the future. The general arguments for this assessment are the following.

- Digital concepts offer several advantages which augment the existing strengths of fluid power systems.
  - High reliability and robustness: switching valves are more robust, for instance with respect to oil contamination; parallel valve technology provides redundancy – a great advantage in reliability and availability and in heavy duty systems.
  - Cost advantage compared with electrical drives: switching valves can probably be manufactured at lower cost than proportional valves; the higher standardization rate of valves in digital systems supports this production cost advantage with reduced cost for service and spare parts.
  - High force and power density: switching valves have the potential to become very compact and light components.
- Digital concepts avoid shortcomings of existing continuous fluid power systems.
  - Simplified systems: software allows versatile and flexible configurations. The same digital hydraulic valve assembly can be used as a directional, flow control or pressure control valve.
  - Low efficiency of resistance control can be improved by several digital concepts, like switching converters, digital cylinders or digital displacement pumps, in particular since these concepts can also recuperate energy.
  - Continuous fluid power concepts do not harmonize with digital electrical drive systems and digital control concepts: digital fluid power concepts integrate into the digital electrical and electronic world.
  - Continuous systems need a proper dimensioning for every application to exploit the control range optimally and this hampers standardization of components: digital concepts use multiple arrangements or control by timing; both concepts are much more accurate and allow for a large range of sizes with a limited number of different components.
- Digital systems offer new functionality that is impossible with existing fluid power systems. Examples of this are as follows.
  - Sensorless incremental actuation: what is known from stepper motors for sensorless incremental positioning can be accomplished with certain switching concepts with fluid power drives; this can be exploited more widely, for instance for the precise dosing of fluid quantities in process technology.
  - A very precise resolution of an intended large control range by parallelization of units (for instance valve units) in combination with appropriate coding schemes, e.g. with binary coding.

None of these advantages are freely available. The main investments which are required are openness to digital concepts, more pronounced system thinking and the use of proper theoretical and experimental methods and tools for system design and optimization. Such methods and tools will become more widely available and the potential benefits and limitations will be more fully recognized in the striving for innovative technical systems. System manufacturers will more readily adopt fluid power technologies wherever they promise advantage.

Digital concepts will become a strong wing of fluid power technology. They will only partly replace existing continuous technology, but will provide new opportunities for fluid power and a powerful solution-set for engineering.

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